V Muhammed Saad Sabeel

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Professional Summary

Electronics & Telecommunication Engineering student with a genuine, deep-rooted interest in electronics and hardware development. A naturally curious tech enthusiast who brings creativity and problem-solving to every project. Always eager to experiment with new technologies, adapt to new challenges, and continuously evolve as an Engineer.

Education

BMS Institute of Technology & Management — Bengaluru, India
Bachelor of Engineering in Electronics & Telecommunication Engineering, CGPA: 9.24 — *2023 – 2027*

Agragami PU College — Bengaluru, India
Class 12 Science, Percentage: 89% — *2022 – 2023*

Projects

Diabreeze: Non-Invasive Biomarker Detection System | *ESP32, Data Processing*

- Developed a portable diagnostic system that extracts and processes real-time breath acetone telemetry as a biomarker to estimate blood glucose levels.
- Engineered an ESP32 microcontroller data pipeline to compute and visualize physiological and environmental metrics instantly.

QNX Deterministic Security Monitor | *C, QNX RTOS, Microkernel Architecture, IPC*

- Designed a split-architecture security solution on a QNX Microkernel using strictly isolated C programs to mitigate hardware side-channel attacks.
- Developed a zero-overhead Mitigation Engine (Priority 63) triggered by native IPCs to instantly preempt and freeze compromised processes mid-instruction, preventing data exfiltration.

IoT Intrusion Detection via Side-Channel Analysis | *Raspberry Pi, I2C, Python, Edge AI*

- Building a non-intrusive edge monitoring system utilizing hardware power side-channel analysis to detect malware, bypassing the need for software logs.
- Engineered an I2C data acquisition pipeline to sample current consumption every 0.2 seconds, generating a custom dataset to train real-time Edge AI classification models.

Smart Switching Using IR Remote and Mobile | *NodeMCU ESP32, IoT, Blynk*

- Developed home automation system with 4-channel relay module for appliance control.
- Implemented dual control modes: IR remote for local control and Blynk IoT platform for mobile access.

Secure Small Office Network Simulation | *Cisco Packet Tracer, VLANs, Routing*

- Designed and simulated a secure multi-department office network with VLAN-based segmentation, isolating Management, Sales, Engineering, and Guest traffic to enforce structured network access policies.
- Configured inter-VLAN routing, DHCP-based automatic IP allocation, and access control policies for segmented network communication.

Extracurricular

Web Development Intern | *Bhisma's Aero Pvt. Ltd., Bengaluru*

- Developed and maintained web applications for aerospace industry client requirements.

AI Model Evaluator | *Outlier*

- Contributed 330+ hours of RLHF and SFT tasks to train enterprise LLMs, evaluating model outputs for factual accuracy and logical consistency.
- Acted as an Oracle-tier grader across Mathematics and multilingual domains, applying strict quality assurance protocols to AI-generated data.

Technical Skills

Programming Languages: C, Python
Tools & Software: Keil uVision, Xilinx Vivado, Synopsys, QNX Neutrino, Cisco Packet Tracer
Embedded & Systems: ESP32, Raspberry Pi, RTOS